Applied Machine Learning - Basic

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Lecture 7

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Data Science and Computation PhD + Master in Bioinformatics
University of Bologna

https://github.com/dbonacorsi/AML-Basic

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Hands-on:

1 - Data prep - load dataset

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NumPy

NumPy

https://numpy.org/

A third-party package added to Python to support scientific computing

- in particular, it provides you with **multi-dimensional array objects**
  - i.e. support matrix manipulation, linear algebra, all operations you might want to do on large collection of numbers (e.g. plenty in ML!)

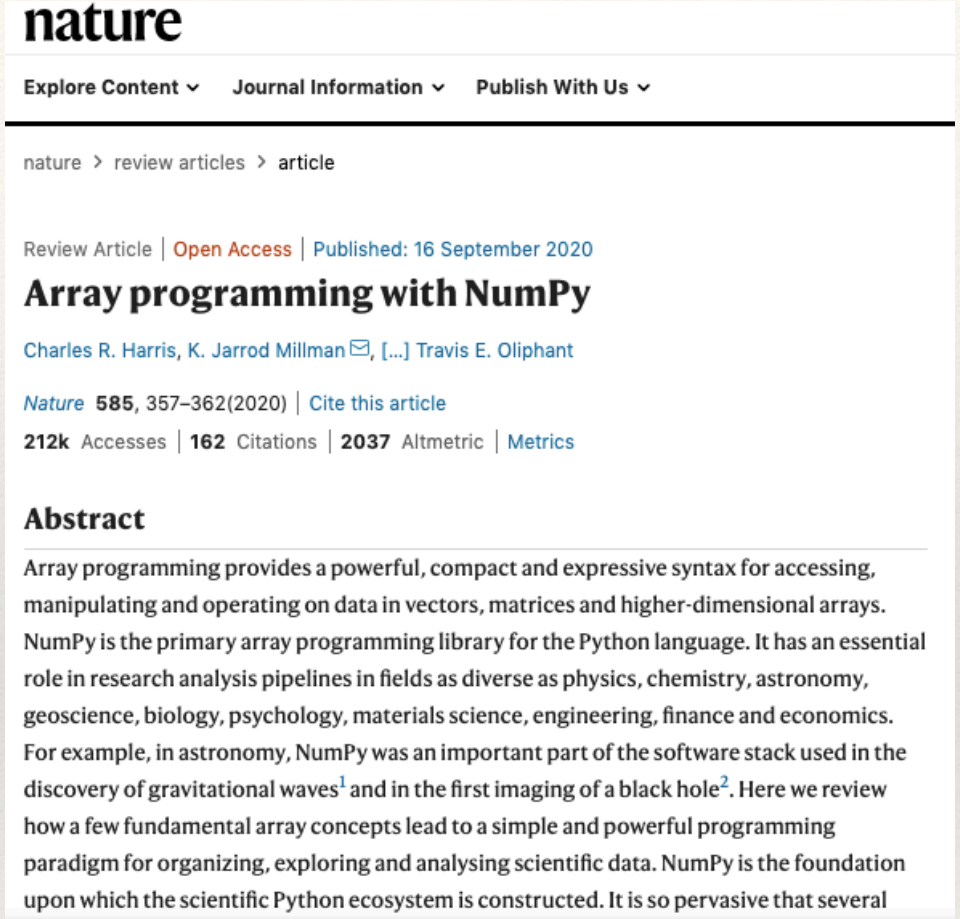
Official NumPy paper

The NumPy paper on **Nature**

- https://www.nature.com/articles/s41586-020-2649-2

To be formally cited as:

- Harris, C.R., Millman, K.J., van der Walt, S.J. *et al.* Array programming with NumPy. *Nature* **585,** 357–362 (2020). https://doi.org/10.1038/s41586-020-2649-2



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Review Article | Open Access | Published: 16 September 2020

Array programming with NumPy

Charles R. Harris, K. Jarrod Millman, [...] Travis E. Oliphant

Nature **585**, 357–362(2020) | Cite this article

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Abstract

Array programming provides a powerful, compact and expressive syntax for accessing, manipulating and operating on data in vectors, matrices and higher-dimensional arrays. NumPy is the primary array programming library for the Python language. It has an essential role in research analysis pipelines in fields as diverse as physics, chemistry, astronomy, geoscience, biology, psychology, materials science, engineering, finance and economics. For example, in astronomy, NumPy was an important part of the software stack used in the discovery of gravitational waves[1] and in the first imaging of a black hole[2]. Here we review how a few fundamental array concepts lead to a simple and powerful programming paradigm for organizing, exploring and analysing scientific data. NumPy is the foundation upon which the scientific Python ecosystem is constructed. It is so pervasive that several

Not to be discussed deeply in this course, but you should be aware of the gender unbalance controversy behind this paper. More details at the lecture.

NumPy 1D array

```
import numpy as np
```

Just a convention, but almost always followed

```
a = np.array( [1, 2, 3] )
```

1-D array object

```
print(a)
```

```
[1 2 3]
```

Printed out as a 1D array

```
print( a[0] )
```

```
1
```

Supports matrix manipulation and linear algebra

NumPy 2D array

We can construct 2D arrays, as multi-dimensional arrays with any number N of dimensions, otherwise known (despite not mathematically precise) as "tensors"

```
import numpy as np

a = np.array( [[1, 2, 3], [4, 5, 6]] )

print(a)
```

a list of lists..

```
[[1 2 3]
 [4 5 6]]
```

2-D array object

Printed out in a form that suggest rows and columns of a matrix. This is intentional: NumPy is intended for matrix manipulation so indeed it effectively understands 2D objects as a matrix and prints it out in rows and columns

```
print( a.ndim )
print( a.size )
print( a.shape )
print( a.dtype )
```

```
2
6
(2, 3)
int32
```

Tuple

a tuple is a read-only list

size = total nb of elements in the arrays

shape = nb elements in each dimension

type: all NumPy array elements must be of the same type (lists instead can contain mixed data types)

NumPy 2D array

```
print(a)
```

```
[[1 2 3]
 [4 5 6]]
```

```
print( a[1, 0] )
print( a[0, 0:3] )
print( a[0, :] )
print( a[:, 0] )
```

```
4
[1 2 3]
[1 2 3]
[1 4]
```

slices

[row, col]
Row
Row
Column

“:” selects all elements

Initialising/building arrays

```
a = np.zeros((2, 3))
```

```
[[ 0.  0.  0.]
 [ 0.  0.  0.]]
```

```
a = np.arange(4)
```

```
[0 1 2 3]
```

```
a = np.arange(1, 25).reshape(2, 3, 4)
```

```
[[[ 1  2  3  4]
  [ 5  6  7  8]
  [ 9 10 11 12]]

 [[13 14 15 16]
  [17 18 19 20]
  [21 22 23 24]]]
```

Elementwise operations

NumPy

```
a = np.arange(9).reshape(3,3)
```

```
[[0 1 2]
 [3 4 5]
 [6 7 8]]
```

```
a + a
```

```
[[ 0  2  4]
 [ 6  8 10]
 [12 14 16]]
```

```
a * a
```

```
[[ 0  1  4]
 [ 9 16 25]
 [36 49 64]]
```

Combining arrays and scalars

NumPy

```
a = np.arange(9).reshape(3,3)
```

```
[[0 1 2]
 [3 4 5]
 [6 7 8]]
```

```
print( a + 1 )
```

add a scalar to EACH element of the matrix

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

```
print( a * 4 + 1 )
```

```
[[ 1  5  9]
 [13 17 21]
 [25 29 33]]
```

Broadcasting

NumPy

NumPy arrays also support broadcasting, a cool feature.

2D array

```
a = np.arange(9).reshape(3,3)
```

a

```
[[0 1 2]
 [3 4 5]
 [6 7 8]]
```

1D array

```
b = np.array([10, 20, 30])
```

b

```
[10 20 30]
```

Matrix + Vector ???

Yes! The elements of the vector are broadcasted to have the same shape of the matrix we are adding them to, and a 3x3 matrix is created

b is broadcast

```
[[10 20 30]
 [10 20 30]
 [10 20 30]]
```

```
print( a + b )
```

```
[[10 11 12]
 [13 14 15]
 [16 17 18]]
```

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- datasets
- images
- AMLBas_1_DataPrep_LoadDataset.ipynb
- AMLBas_2_DataPrep_ExploreData_part1.ipynb
- AMLBas_3_DataPrep_ExploreData_part2.ipy...
- AMLBas_4_DataPrep_PrepareData.ipynb
- AMLBas_5_DataPrep_FeatSelect.ipynb
- AMLBas_6_DataPrep_EvalResampling.ipynb
- AMLBas_7_Modelling_AlgoPerfMetrics.ipynb
- AMLBas_8_Modelling_CompareModels.ipynb
- AMLBas_9_Modelling_ImprovePerfAlgoTunin...
- README.md

Look into these:

pima-indians-diabetes.data.csv

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Hands-on:

2 - Data prep - explore data (part 1)

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Hands-on:

3 - Data prep - explore data (part 2)

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Hands-on:

4 - Data prep - Prepare data

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Hands-on:

5 - Data prep - Feature selection

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